JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Department of
Computer Science & Engineering

B. Tech. in
Computer Science & Engineering
V SEM /3rd Year
Syllabus: 2026-27

Education to Eternity

।।ज्ञानम् सर्वार्थ साधनम्।।

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Program Name: B. Tech. Computer Science & Engineering **Program Code: CS**

Semester V

Sr. No	Category of Subject	Course Code	Course Name	Teaching Scheme			Evaluation Scheme				Credit	Teaching Mode
				L	T/A	P	CA	MSE	ESE	Total		
1	PCC	CS5T001	Database Management System	2	0	0	20	20	60	100	2	Class Board /PPT
2	PCC	CS5T002	Machine Learning	2	0	0	20	20	60	100	2	Class Board /PPT
3	PCC	CS5T003	Internet of Things	2	0	0	20	20	60	100	2	Class Board /PPT
4	PEC	CS5E001	PEC-I	2	0	0	20	20	60	100	3	Class Board /PPT
5	PEC	CS5E002	PEC-II	3	0	0	20	20	60	100	3	Class Board /PPT
6	MDM	CS5M003	MDM-III	2	0	0	20	20	60	100	2	Class Board /PPT
7	MDM	CS5M004	MDM-IV	2	0	0	20	20	60	100	2	Class Board /PPT
8	OEC	CS5O003	Open Elective -III	2	0	0	20	20	60	100	2	Class Board /PPT
9	PCC	CS5L001	Database Management System Lab	0	0	2	60	-	40	100	1	PCs
10	PCC	CS5L002	Machine Learning Lab	0	0	2	60	-	40	100	1	PCs
11	ELC	CS5P003	Field Project/ Mini Project-III	0	0	2	60	-	40	100	1	PCs
12	ELC	CS5I001	Internship-I	0	0	4	60	-	40	100	1	PCs
13	Audit	CS5T004	JDCAP-III: Placement Preparation Training	2	0	0	30	-	20	50	Audit	Class Board /PPT
			Total	**19**	**0**	**10**	**430**	**160**	**660**	**1250**	**22**	

Version of the Scheme	JD/ACD/CSE/2024-28/3	Signatures of all BOS members
Year of release of the scheme document:	2026-27	

Secretary BOS

Chairman BOS

Deputy Dean (Acd)

Dean (Acd)

Dean (IQAC)

Chairman (Acd.Council)

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Professional Elective Courses (PEC) List

Sr. No.	Subject Code	CSE Domain				
		PEC Category	Database	Computing Domain	Computer Vision Domain	Systems
1	CS5E001	PEC-I	Blockchain Technology	Cluster Computing	Interactive Graphics	Transmission Control System
2	CS5E002	PEC-II	Big Data Analytic Tools & Techniques	Distributed Computing	React JS	Computer Forensic
3	CS3O003	OEC-III	Advanced Business Application Programming			
4	CS5M003	MDM-III	Prompt Engineering			
5	CS5M004	MDM-IV	Ethics, Bias & Safety in LLM			

Version of the Scheme	JD/ACD/CSE/2024-28/2	Signatures of all BOS members
Year of release of the scheme document:	2025-26	

Secretary BOS

Chairman BOS

Deputy Dean (Acd)

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Dean (IQAC)

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS5T001**	**Database Management Systems**	**2**	**0**	**0**	**2**

Prerequisites for the course	
1.	Basics of Data, Database, Data Management Tools.
2.	Fundamental of File, File Organization etc.

Prior Reading Material/useful links	
1.	https://onlinecourses.nptel.ac.in/noc23_cs29/preview
2.	https://www.techtarget.com/searchdatamanagement/definition/database-management-system
3.	https://www.youtube.com/watch?v=c5HAwKX-suM

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to understand fundamentals of database management system.
2	CO2	Student will be able to classify the different query development knowledge.
3	CO3	Student will be able to apply modelling and normalization to databases.
4	CO4	Student will be able to analyse query processing and optimization techniques.
5	CO5	Student will be able to evaluate the File Organization, Indexing and exhibit the knowledge of transaction and concurrency control.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Database Systems:** Significance and advantages, Types of Databases, Limitations of File processing system, the DBMS Environment, Data Abstraction, Data Independence, Data Definition Language (DDL), Data Manipulation Language (DML).	**05**

	Data models: Evolution of Data Models, Entity-relationship model, Relational integrity constraints, data manipulation operations.	
Unit II	**Relational query languages:** Relational algebra, Tuple and domain relational calculus,SQL3, DDL and DML constructs, Open source and Commercial DBMS – MYSQL,ORACLE, DB2, SQL server.	**05**
Unit III	**Relational database design:** Normalization of Database Tables: Need and Significance, Domain and data dependency, Armstrong's axioms, Normal forms, Dependency preservation, Lossless design.	**05**
Unit IV	**Query processing:** Evaluation of relational algebra expressions, Query equivalence, Join strategies.	**05**
Unit V	**File Organization and Indexing:** Indices, B-trees, hashing. **Transaction processing:** Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp based schedulers, Multi-version and optimistic Concurrency Control schemes, Database recovery.	**06**

Text Books	
1.	Henry Korth, Abraham Silberschatz& S. Sudarshan, Database System Concepts, McGraw-Hill Publication, 6th Edition, 2011.
2.	Bipin Desai, An Introduction to Database System, West Publishing Company, College & School Division, 1990.
3.	RaghuRamakrishnan, Johannes Gehrke, Database Management Systems, McGraw-Hill Publication, 3rd Edition, 2003.

Reference Books	
1.	Joel Murach, Murach's Oracle SQL and PL/SQL for Developers, Mike Murach& Associates, 2nd Edition, 2014.
2.	Wiederhold, *Database Design*, McGraw-Hill Publication, 2nd Edition, 1983.
3.	Navathe, Fundamentals of Database System, Addison-Wesley Publication, 6 th Edition, 2012.

Useful Links	
1.	https://www.youtube.com/watch?v=T7AxM7Vqvaw
2.	https://www.guru99.com/what-is-dbms.html
3.	https://nptel.ac.in/courses/106105175

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5T002	Machine Learning	2	0	0	2

Prerequisites for the course	
1.	Understanding how to load, clean, manipulate, and explore data is crucial.
2.	Basics of common algorithms and data structures.

Prior Reading Material/useful links	
1.	https://www.javatpoint.com/machine-learning
2.	https://www.tutorialspoint.com/machine_learning/index.htm
3.	https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to understand the basics of machine learning and its components.
2	CO2	Students will be able to explain the decision tree learning algorithms to classify multivariate data.
3	CO3	Students will be able to interpret the Bayesian Learning for finding relationships between data variables.
4	CO4	Students will be able to develop and apply Linear Models for Regression algorithms for learning to control complex systems.
5	CO5	Students will be able to write scientific reports on computational machine learning methods, results and conclusions.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Well-posed learning problems, designing a Learning System, Perspectives and Issues in Machine learning, **Concept Learning and General-to-specific Ordering:** A concept learning task, Concept learning as Search, Finding a maximally specific hypothesis, Version Spaces and Candidate elimination algorithm, Inductive Bias.	05

Unit II	**Unsupervised Learning:** Clustering Learning from unclassified data, Hierarchical Aglomerative Clustering, k-means partitional clustering, Batchler and Wilkin's algorithm, Study of Supervised learning and Reinforcement Learning.	**05**
Unit III	**Bayesian Learning:** Bayes theorem and concept learning, Maximum likelihood and least square error hypotheses, Minimum description length principle, Bayes optimal classifier, Gibb's algorithm, Naive Bayes classifier, **Computational Learning Theory:** Probably learning an approximately correct hypothesis, PAC learnability, The VC dimension, the mistake bound model for learning.	**05**
Unit IV	**Linear Models for Regression:** Linear basis function models, The Bias-Variance decomposition, Bayesian Linear Regression, Bayesian Model comparison, **Kernel Methods:** Constructing kernels, Radial basis function networks, Gaussian Processes, **Ensemble Learning:** Bagging, boosting, and DECORATE. Active learning with ensembles.	**06**
Unit V	**Decision Tree Learning:** Decision tree learning algorithm, Hypothesis space search in decision tree **Evaluating Hypothesis:** Estimating Hypothesis accuracy, Basics of sampling theory, deriving confidence intervals, Hypothesis testing, comparing learning algorithms.	**05**

Text Books	
1.	Tom M. Mitchell, "Machine Learning", McGraw-Hill, 2010
2.	Introduction to Machine Learning, Fourth Edition By Ethem Alpaydin-2020-MIT Press

Reference Books	
1.	Ethem Alpaydin, (2004) "Introduction to Machine Learning (Adaptive Computation and Machine Learning)", The MIT Press.
2.	T. astie, R. Tibshirani, J. H. Friedman, "The Elements of Statistical Learning", Springer(2nd ed.), 2009

Useful Links	
1	https://alex.smola.org/drafts/thebook.pdf
2	https://www.google.co.in/books/edition/An_Introduction_to_Machine_Learning/q6UzDwAAQBAJ?hl=en&gbpv=1&dq=introduction+to+machine+learning&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5T003	Internet of Things	2	0	0	2

Prerequisites for the course	
1.	Knowledge of Computer networks, LAN, WAN & MAN
2.	Study of Protocols, Sensors, basics of IoT.

Prior Reading Material/useful links	
1.	https://www.youtube.com/watch?v=6mBO2vqLv38
2.	https://www.internetsociety.org/iot/
3.	https://www.forbes.com/sites/bernardmarr/2021/12/13/the-5-biggest-internet-of-things-iot-trends-in-2022/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Identify the different technology.
2	CO2	Student will able to Apply IoT to different applications.
3	CO3	Student will able to Analysis and evaluate protocols used in IoT.
4	CO4	Student will able to Design and develop smart city in IoT.
5	CO5	Student will able to Analysis and evaluate the data received through sensors in IoT.

Syllabus:

Course Contents		Hours
Unit I	**IoT Introduction:** Origin of IoT, IoT and Digitization, IoT Impact, Convergence of IT and IoT, IoT challenges, Need of IoT, IoT features, Bulding blocks of IoT, IoT Network Architecture and Design, Drivers Behind New Network Architectures, Comparing IoT Architectures, The Core IoT Functional Stack, IoT Data Management and Compute Stack. IoT Things : Sensors and Actuators	**06**
Unit II	**IoT Ecosystem:** Three layered architecture, five-layer architecture, cloud computing, fog computing, IoT taxonomy. Connectivity Terminology: IoT LAN , IoT WAN, IoT Node, IoT Gateway.	**04**

Unit III	**IoT Protocols:** IoT Networking protocols: MQTT, SMQTT, CoAP, XMPP, AMQP. IoT Communication protocols : IEEE 802.15.4, Zigbee, 6LoWPAN, Wireless HART, Z-Wave, Bluetooth, NFC, RFID.	**04**
Unit IV	**Data Analytics for IoT:** An Introduction to Data Analytics for IoT, Machine Learning, Big Data Analytics Tools and Technology, Edge Streaming Analytics, Network Analytics, Securing IoT, A Brief History of IoT Security, Common Challenges in IoT Security, Formal Risk Analysis Structures: OCTAVE and FAIR, The Phased Application of Security in an Operational Environment.	**06**
Unit V	**Implementation of IoT with Arduino and RaspberryPi:** Introduction to Arduino, Integration of sensors and actuators with Arduino, IDE programming, Introduction to RaspberryPi, About the RaspberryPi Board: Hardware Layout, Operating Systems on RaspberryPi, Configuring RaspberryPi, Programming RaspberryPi with Python, Application of IOT.	**06**

Text Books	
1.	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry,"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", 1 st Edition, Pearson Education
2.	Srinivasa K G, "Internet of Things", CENGAGE Leaning India, 2017.
3.	Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on-Approach)", 1st Edition, VPT,2014.

Reference Books	
1.	Raj Kamal, "Internet of Things: Architecture and Design Principles", 1st Edition, McGraw Hill Education, 2017.
2.	Analytics for the Internet of Things (IoT): Intelligent analytics for your intelligent devices",by Andrew Minteer
3.	"Internet of Things: Architectures, Protocols and Standards", by Simone Cirani, Gianluigi Ferrari, Marco Picone, and Luca Veltri

Useful Links	
1.	https://www.techtarget.com/iotagenda/definition/Internet-of-Things-IoT
2.	https://builtin.com/internet-things
3.	https://www.simplilearn.com/iot-devices-article

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS5E001A**	**PEC-I (Block Chain Technology)**	2	0	0	**2**

Prerequisites for the course	
1.	Basic knowledge of Blockchain Technology with uses & applications.
2.	Basics of Operating Systems, Cryptography and Network Security.

Prior Reading Material/useful links	
1.	https://onlinecourses.nptel.ac.in/noc22_cs44/preview
2.	https://nptel.ac.in/courses/106105235
3.	https://www.ibm.com/topics/blockchain

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will able to Understand emerging abstract models for Blockchain Technology.
2	CO2	Student will able to Identify major research challenges and technical gaps existing between theory and practice in crypto currency domain.
3	CO3	Student will able to provides conceptual understanding of the function of Blockchain as a method of securing distributed ledgers, how consensus on their contents is achieved, and the new applications that they enable.
4	CO4	Student will able to Apply hyper ledger Fabric and Etherum platform to implement the Block chain Application
5	CO5	Student will able to design applications based on blockchain technology for E-Governance, Land Registration, Medical Information Systems, and others

Syllabus:

	Course Contents	Hours
Unit I	**Introduction:** Blockchain-History, Myths, Benefits, Limitations and Challenges of Blockchain, Structure of Blocks, Miners, Working of Blockchain, Types of Blockchain, Blockchain as Public Ledgers-Bitcoin, Blockchain 2.0, Smart Contracts, Transactions-Distributed	**09**

	Consensus, The Chain and the Longest Chain -Cryptocurrency to Blockchain 2.0 - Permissioned Model of Blockchain,	
Unit II	**Blockchain Architecture and Cryptographic:** Crypto Primitives, Permissioned Blockchain, Consensus mechanism, Cryptographic, Hash Function, Properties of a hash function-pointer and Merkle tree. Public key cryptosystems, private vs public blockchain. Introduction to cryptographic concepts required, Hashing, public key cryptosystems, private vs public blockchain and use cases,	**07**
Unit III	**Bitcoin Consensus:** Introduction to Bitcoin Blockchain, Transactions, Bitcoin limitations, Bitcoin Consensus, Proof of Work (PoW)- Hashcash PoW , Bitcoin PoW, Attacks on PoW ,monopoly problem- Proof of Stake-Proof of Burn - Proof of Elapsed Time - Bitcoin Miner, Mining Difficulty, Mining Pool-Permissioned model and use cases.	**06**
Unit IV	**Cryptocurrency and Smart Contracts:** Introduction, Ethereum blockchain, Elements of the Ethereum blockchain, IOTA, Namecoin. Legal Aspects Cryptocurrency Exchange, Black Market and Global Economy. Smart Contracts: Definition, DAO, Ricardian contracts, Precompiled contracts.	**06**
Unit V	**Hyperledger Fabric:** Architecture of Hyperledger fabric v1.1- Introduction to hyperledger fabric v1.1, chain code- Ethereum: Ethereum network, EVM, Transaction fee, Mist Browser, Ether, Gas, Solidity, Truffle Design and issue Crypto currency, Mining, DApps, DAO **Blockchain Applications :** Uses of Blockchain in E-Governance, Land Registration, Medical Information Systems, Finance, and others	**08**

Text Books	
1.	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016
2.	Draft version of "S. Shukla, M. Dhawan, S. Sharma, S. Venkatesan, 'Blockchain Technology: Cryptocurrency and Applications', Oxford University Press, 2019.
3.	Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform, 2017.
4.	Mastering Blockchain - Distributed ledgers, decentralization and smart contracts explained, Author- Imran Bashir, Packt Publishing Ltd, Second Edition, ISBN 978-1- 78712-544-5, 2017

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Reference Books	
1.	Nicola Atzei, Massimo Bartoletti, and TizianaCimoli, A survey of attacks on Ethereum smart contracts
2.	Joseph Bonneau et al, SoK: Research perspectives and challenges for Bitcoin and cryptocurrency, IEEE Symposium on security and Privacy, 2015.
3.	Nakamoto, Satoshi, Bitcoin: A peer-to-peer electronic cash system, Research Paper

Useful Links	
1.	https://blockgeeks.com/guides/what-is-blockchain-technology/
2.	https://www.investopedia.com/terms/b/blockchain.asp
3.	https://nptel.ac.in/courses/106106138
4.	https://nptel.ac.in/courses/106104220

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E001B	PEC-I (Cluster Computing)	3	0	0	3

Prerequisites for the course	
1.	Understanding of CPU, memory hierarchy, cache, and instruction-level parallelism.
2.	Familiarity with distributed systems.

Prior Reading Material/useful links	
1.	https://www.esds.co.in/blog/cluster-computing-definition-architecture-and-algorithms/
2.	https://www.tutorialspoint.com/what-is-cluster-computing

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to understand the concepts of cluster computing, and differentiate it from grid and cloud computing models.
2	CO2	Student will be able to Identify and analyse cluster hardware components, architectures, and resource management techniques for effective system design. .
3	CO3	Student will be able to Demonstrate the ability to plan, configure, and administer cluster environments, including network setup, monitoring, and job scheduling.
4	CO4	Student will be able to Apply parallel programming models such as MPI and Open MP to develop efficient distributed applications on cluster systems.
5	CO5	Student will be able to Explore advanced topics in cluster computing, including HPC, GPU acceleration, distributed file systems, and cloud-based cluster integration.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Cluster Computing: Definitions and motivations, Types of parallel computers, Cluster vs. Grid vs. Cloud computing, Applications of cluster computing, Benefits and challenges.	**08**

Unit II	**Cluster Architecture:** Cluster hardware: nodes, interconnects, and network topology, Cluster operating systems, Middleware and resource management, High availability and fault tolerance, Load balancing strategies.	**08**
Unit III	**Cluster Setup and Administration:** Planning and designing a cluster, Hardware and software configuration, Network setup and storage management, Cluster monitoring tools (e.g., Ganglia, Nagios), Job scheduling tools (e.g., PBS, SLURM).	**08**
Unit IV	**Programming Models and Tools:** MPI (Message Passing Interface), OpenMP for shared memory programming, Hybrid programming approaches, Remote Procedure Calls (RPCs), Debugging and profiling tools.	**06**
Unit V	**Case studies and research trends:** High Performance Computing (HPC), GPU clusters and accelerators, Distributed file systems (e.g., HDFS, Lustre) Cloud-based clusters (e.g., AWS, Google Cloud)	**06**

Text Books	
1.	**Rajkumar Buyya** – High Performance Cluster Computing, Vol I & II.
2.	Thomas Rauber & Gudula Rünger – Parallel Programming: for Multicore and Cluster Systems

Reference Books	
1.	High Performance Cluster Computing: Architectures and Systems, Vol. 1, Prentice Hall
2.	Grid and Cluster Computing, Prabhu C.S.R, PHI Learning Private Limited
3.	In search of clusters (2nd ed.), Gregory F. Pfister, IBM, Austin, TX, Prentice Hall

Useful Links	
1.	https://www.techtarget.com/whatis/definition/cluster
2.	https://www.studocu.com/in/document/mahatma-gandhi-university/bachelor-of-computer-applications/clusters-lecture-notes-cloud-computing/20905472

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E001C	PEC-I (Interactive Graphics)	3	0	0	3

Prerequisites for the course	
1.	Basics of Image, Graphics, Audio, Video etc.
2.	Generalize the knowledge about Interactive Computer Graphics's architecture, goals, etc .

Prior Reading Material/useful links	
1.	https://www.javatpoint.com/interactive-and-passive-graphics
2.	https://www.coursera.org/learn/interactive-computer-graphics
3.	https://www.youtube.com/watch?v=UVCuWQV_-Es

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to understand the basic concepts of display techniques in computing.
2	CO2	Student will be able to analyse different display code generation and Graphical functions
3	CO3	Student will be able to interpret the Interactive graphical techniques
4	CO4	Student will be able to apply Two-dimensional Geometric Transformations.
5	CO5	Student will be able to develop 3-dimesional Transformations. Interactive Graphical Techniques GUI.

Syllabus:

Course Contents		Hours
Unit I	**Display Devices:** Line and point plotting systems: Raster, Vector, pixel and point plotters, Continual refresh and storage displays, Digital frame buffer, Plasma panel display, Very high resolution devices. High-speed drawing. Display processors. Character generators, Colour Display techniques (shadowmask and penetration CRT, colour look-up tables, analog false colours, hard copy colour printers).	**09**

Unit II	**Display Description:** Screen co-ordinates, user co-ordinates, Graphical data structures (compressed incremental list, vector list, use of homogeneous coordinates); Display code generation Graphical functions: the view algorithm. Two-dimensional transformation, Line drawing. Circle drawing algorithms.	**06**
Unit III	**Interactive graphics:** Pointing and positing devices (cursor, lightpen, digitizing tablet, the mouse, track balls). Interactive graphical techniques. Positioning (Elastic or Rubber Bank lines, Linking, zooming, panning clipping, windowing, scissoring). Mouse Programming.	**06**
Unit IV	**Two-dimensional Geometric Transformations: Basic** Transformations, Matrix Representation and Homogeneous Coordinates, Composite Transformations, Reflection and Shearing. Two-Dimension Viewing: The viewing Pipeline, Window to view port coordinate transformation, Clipping Operations, Point Clipping, Line Clipping, Polygon Clipping, Text Clipping, Exterior Clipping	**07**
Unit V	**3-D Graphics:** Wire-frame, perspective display, Perspective depth, projective transformations, Hidden line and surface elimination. Transparent solids, shading, Two dimensional Transformations. 3-dimesional Transformations. Interactive Graphical Techniques GUI.	**08**

Text Books	
1.	Foley & Van Dam : Fundamentals of Interactive Computer Graphics, Addison- Wesley.
2.	Newman : Principles of Interactive Computer Graphics, McGraw Hill.
3.	Introduction to interactive computer graphics By Joan E. Scott -1982

Reference Books	
1.	Donald Hearn & M. Pauline Baker, "Computer Graphics with OpenGL", Third Edition, 2004, Pearson Education, Inc. New Delhi
2.	Tosijasu, L.K. : Computer Graphics, Springer-Verleg.
3.	Jeffcoate : Multimedia in Practice, Prectice-Hall.

Useful Links	
1	https://edutechlearners.com/interactive-computer-graphics-icg-notes/
2	https://www.google.co.in/books/edition/Essentials_of_Interactive_Computer_Graph/PqT3RRVo4isC?hl=en&gbpv=1&dq=interactive+graphics&printsec=frontcover
3	https://graphics.cs.utah.edu/courses/cs6610/spring2022/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E001D	PEC-I (Transmission Control System)	3	0	0	3

Prerequisites for the course	
1.	Understand the principles of networking.
2.	Study the basics of protocol.

Prior Reading Material/useful links	
1.	https://www.techtarget.com/searchnetworking/definition/TCP-IP
2.	https://www.youtube.com/watch?v=GfaHdjApnhU
3.	https://www.techtarget.com/searchnetworking/definition/TCP

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understand Network Models and Protocol Layers.
2	CO2	Student will be able to Analyze TCP Header and Connection Management Mechanisms.
3	CO3	Student will be able to Apply Flow Control and Error Handling Techniques.
4	CO4	Student will be able to Evaluate Congestion Control Algorithms and TCP Variants.
5	CO5	Student will be able to Develop TCP-Based Network Applications and Analyze Traffic.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Overview of network architectures, TCP/IP Stack, OSI and TCP/IP models, Role of protocols in networking, Transport Layer, Functions of the transport layer, Role of Transport Layer: TCP vs UDP, Services Provided by TCP, TCP Header Format and Fields, Port Numbers and Multiplexing/Demultiplexing.	7

Unit II	**TCP Connection Management and State Transitions:** TCP Connection Establishment: Three-Way Handshake, TCP Connection Termination: Four-Way Handshake, TCP States and State Diagram, Simultaneous Open/Close and Half-Closed Connections, TCP Timers: Retransmission, Persistence, Keep-alive, TIME_WAIT	**7**
Unit III	**Flow Control and Error Recovery:** Sliding Window Protocol, Receive/Send Buffers, Silly Window Syndrome (SWS) and Nagle's Algorithm, Acknowledgment Strategies: Cumulative ACK, Delayed ACK, Error Detection using Checksum, Timeout and Retransmissions, Retransmission Algorithms: Go-Back-N, Selective Repeat, SACK	**7**
Unit IV	**Congestion Control and Performance:** Causes of Network Congestion, Congestion Control Algorithms: Slow Start, Congestion Avoidance, Fast Retransmit and Fast Recovery, TCP Variants: Reno, Tahoe, Vegas (brief overview), TCP Performance Tuning and Optimization Techniques, TCP Over Wireless and Mobile Networks.	**7**
Unit V	**Applications, Socket Programming & Analysis Tools:** TCP Socket Programming Basics (C/Python), Client-Server Communication over TCP, Tools: Wireshark, TCPdump, Netstat, ss command, Packet Sniffing and TCP Flow Analysis, TCP Enhancements and Emerging Protocols: TCP Options (Window Scaling, Timestamps), Introduction to QUIC (vs TCP)	**8**

Text Books	
1.	*Computer Networks: A Systems Approach* by Larry L. Peterson and Bruce S. Davie
2.	*Computer Networking: Principles, Protocols and Practice* by Olivier Bonaventure
3.	*TCP/IP Illustrated* by W. Richard Stevens

Reference Books	
1.	IPv6 Theory, Protocol and Practice, Pete Loshin, 2nd edition, Morgan Kaufmann, 2003.
2.	Internetworking TCP/IP, Comer D.E and Stevens D.L, Volume III, Prentice Hall of India, 1997.

Useful Links	
1.	https://nptel.ac.in/courses/106105183
2.	https://archive.nptel.ac.in/courses/106/105/106105081/

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E002A	PEC-II (Big Data Analytic Tools & Technique)	3	0	0	3

Prerequisites for the course	
1.	Understand the key issues in big data management
2.	Study the tools and techniques required in handling large amounts of datasets.

Prior Reading Material/useful links	
1.	https://sist.sathyabama.ac.in/sist_coursematerial/uploads/SIT1606.pdf
2.	https://www.getsmarter.com/blog/career-advice/big-data-analysis-techniques/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Understand the key issues in big data management and its associated applications in intelligent business and scientific computing.
2	CO2	Student will able to Acquire fundamental enabling tools and techniques and scalable algorithms like Hadoop, Map Reduce and NO SQL in big data analytics
3	CO3	Student will able to Interpret business models and scientific computing paradigms, and apply software tools for big data analytics.
4	CO4	Student will able to Achieve adequate perspectives of big data analytics in various applications like recommender systems, social media applications etc.
5	CO5	Student will able to Develop Big Data Solutions using Hadoop Eco System

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Big Data:** Introduction to Big Introduction to Big Data, Big Data characteristics, types From 0-3 Data of Big Data,	7

	Traditional vs. Big Data business approach, Ref. Case Study of Big Data Solutions.	
Unit II	**Introduction to Hadoop** : What is Hadoop? Core Hadoop Components; Hadoop Ecosystem; Physical Architecture; Hadoop limitations.	6
Unit III	**NoSQL:** What is NoSQL? NoSQL business drivers, NoSQL case studies, NoSQL data architecture patterns: Key-value stores, Graph stores, Column family (Bigtable) stores, Document stores, Variations of NoSQL architectural patterns; Using NoSQL to manage big data: What is a big data NoSQL solution? Understanding the types of big data problems; analyzing big data with a shared-nothing architecture; choosing distribution models: master-slave versus peer-to-peer; four ways that NoSQL systems handle big data problems	7
Unit IV	**Map Reduce and the New Software:** Distributed File Systems: Physical Organization of Compute Nodes, Large-Scale File-System Organization. MapReduce: The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, Details of MapReduce Execution, Coping With Node Failures. Algorithms Using MapReduce: Matrix-Vector Multiplication by MapReduce, Relational-Algebra Operations, Computing Selections by MapReduce, Computing Projections by MapReduce, Union, Intersection, and Difference by MapReduce, Computing Natural Join by MapReduce, Grouping and Aggregation by MapReduce, Matrix Multiplication, Matrix Multiplication with One MapReduce Step.	10
Unit V	**Finding Similar Item:** Applications of Near-Neighbor Search, Jaccard Similarity of Sets, Similarity of Documents, Collaborative Filtering as a Similar-Sets Problem .	6

Text Books	
1.	Anand Rajaraman and Jeff Ullman "Mining of Massive Datasets", Cambridge University Press,
2.	Dan McCreary and Ann Kelly "Making Sense of NoSQL" – A guide for managers and the rest of us, Manning Press.
3.	Alex Holmes "Hadoop in Practice", Manning Press, Dreamtech Press.
4.	Bill Franks , "Taming The Big Data Tidal Wave: Finding Opportunities In Huge Data Streams With Advanced Analytics", Wiley

Reference Books	
1.	Chuck Lam, "Hadoop in Action", Dreamtech Press
2.	Judith Hurwitz, Alan Nugent, Dr. Fern Halper, Marcia Kaufman, "Big Data for Dummies", Wiley India
3.	Paul Zikopoulos, Chris Eaton, "Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data', McGraw Hill Education.
4.	Boris Lublinsky, Kevin T. Smith, Alexey Yakubovich, "Professional Hadoop Solutions", Wiley India.

Useful Links	
1.	https://www.aalimec.ac.in/wpcontent/uploads/2019/09/MC5502_BDA_UNIT_I_NOTES.pdf
2.	https://www.youtube.com/watch?v=pkPdhznqEI4
3.	https://www.iare.ac.in/sites/default/files/NEW%20LECHURE%20NOTES.pdf
4.	https://www.google.co.in/books/edition/Developing_Analytic_Talent/tp46AwAAQBAJ?hl=en&gbpv=1&dq=big+data+analytics+techniques+COURSE+OUTCOMES&printsec=frontcover

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E002B	PEC-II (Distributed Computing)	3	0	0	3

Prerequisites for the course	
1.	Knowledge of computation and communication models.
2.	Basics of agreement protocols.

Prior Reading Material/useful links	
1.	https://www.aalimec.ac.in/wp-content/uploads/Material/cse/3/CS3551%20%20Distributed%20Computing.pdf
2.	https://www.studocu.com/in/document/anna-university/distributed-computing/cs3551-distributed-computing-syllabus/99836399
3.	https://archive.mu.ac.in/myweb_test/MCA%20study%20material/M.C.A.(Sem%20-%20V)%20Distributed%20Computing.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to understand the basic concepts of foundations of distributed systems.
2	CO2	Student will able to Solve synchronization and state consistency problems.
3	CO3	Student will able to Use resource sharing techniques in distributed systems.
4	CO4	Student will able to Apply working model of consensus and reliability of distributed systems.
5	CO5	Student will able to Explain the fundamentals of cloud computing.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Definition-Relation to Computer System Components – Motivation – Message -Passing Systems versus Shared Memory Systems – Primitives for Distributed Communication – Synchronous versus Asynchronous Executions – Design Issues and Challenges; A Model of Distributed Computations: A Distributed Program – A	**06**

	Model of Distributed Executions – Models of Communication Networks – Global State of a Distributed System.	
Unit II	**Logical Time and Global State:** Logical Time: Physical Clock Synchronization: NTP – A Framework for a System of Logical Clocks – Scalar Time – Vector Time; Message Ordering and Group Communication: Message Ordering Paradigms – Asynchronous Execution with Synchronous Communication – Synchronous Program Order on Asynchronous System – Group Communication – Causal Order – Total Order; Global State and Snapshot Recording Algorithms: Introduction – System Model and Definitions – Snapshot Algorithms for FIFO Channels	**09**
Unit III	**Distributed Mutex and Deadlock:** Distributed Mutual exclusion Algorithms: Introduction – Preliminaries – Lamport's algorithm – RicartAgrawala's Algorithm –– Token-Based Algorithms – Suzuki-Kasami's Broadcast Algorithm; Deadlock Detection in Distributed Systems: Introduction – System Model – Preliminaries – Models of Deadlocks – Chandy-Misra-Haas Algorithm for the AND model and OR Model.	**09**
Unit IV	**Consensus:** Consensus and Agreement Algorithms: Problem Definition – Overview of Results – Agreement in a Failure-Free System(Synchronous and Asynchronous) – Agreement in Synchronous Systems with Failures; Checkpointing and Rollback	**06**
Unit V	**Recovery:** Introduction – Background and Definitions – Issues in Failure Recovery – Checkpoint-based Recovery – Coordinated Checkpointing Algorithm – – Algorithm for Asynchronous Checkpointing and Recovery	**05**

Text Books	
1.	Kshemkalyani Ajay D, Mukesh Singhal, "Distributed Computing: Principles, Algorithmsand Systems", Cambridge Press, 2011
2.	Mukesh Singhal, Niranjan G Shivaratri, "Advanced Concepts in Operating systems", McGraw Hill Publishers, 1994
3.	George Coulouris, Jean Dollimore, Time Kindberg, "Distributed Systems Concepts andDesign", Fifth Edition, Pearson Education, 2012

Reference Books	
1.	Pradeep L Sinha, "Distributed Operating Systems: Concepts and Design", Prentice Hall ofIndia, 2007
2.	Tanenbaum A S, Van Steen M, "Distributed Systems: Principles and Paradigms", PearsonEducation, 2007.

Useful Links	
1.	https://www.bbau.ac.in/dept/uiet/CS/IV%20Year%20CS.pdf
2.	https://prathyusha.edu.in/wp-content/uploads/2023/11/DISTRIBUTED-COMPUTING.pdf
3.	https://www.studocu.com/in/document/anna-university/distributed-computing/cs3551-distributed-computing-syllabus/99836399

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E002C	PEC-II (React JS)	3	0	0	3

Prerequisites for the course	
1.	Basics of Java Programming
2.	Fundamental of AngularJS

Prior Reading Material/useful links	
1.	https://www.w3schools.com/angular/
2.	https://www.tutorialspoint.com/angularjs/index.htm
3.	https://www.javatpoint.com/angularjs-tutorial

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to Understand fundamentals of React JS and component-based architecture.
2	CO2	Student will be able to Develop user interfaces using JSX and React components.
3	CO3	Student will be able to Implement state management and event handling in React applications.
4	CO4	Student will be able to Use React Router and APIs to build dynamic web applications..
5	CO5	Student will be able to Develop and deploy responsive single-page React applications.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to React JS:** Introduction to Frontend Frameworks, Basics of React JS, Features of React, Installation & Setup, JSX Syntax, Virtual DOM**,** Components and Props	7
Unit II	**React Components & State:** Functional Components, Class Components, State and Props, Event Handling, Conditional Rendering, Lists and Keys, Forms in React	8

Unit III	**React Hooks:** Introduction to Hooks, useState Hook, useEffect Hook, useContext Hook, Custom Hooks, Component Lifecycle.	8
Unit IV	**Routing & API Integration:** React Router, Navigation and Dynamic Routing, Fetch API / Axios, Handling JSON Data, CRUD Operations, Error Handling	7
Unit V	**Advanced React & Deployment:** Context API, Redux Basics, Performance Optimization, Responsive UI Design, Testing Basics Build & Deployment of React Applications	6

Text Books	
1.	Learning React – Alex Banks & Eve Porcello · 2015
2.	React Up & Running – Stoyan Stefanov · 2013
3.	Ng-book The Complete Book on AngularJS By Ari Lerner · 2013

Reference Books	
1.	Fullstack React – Accomazzo et al. · 2013
2.	Beginning AngularJS By Andrew Grant · 2014
3.	Pro AngularJS By Adam Freeman · 2014

Useful Links	
1.	https://www.w3schools.com/angular/angular_application.asp
2.	https://www.guru99.com/angularjs-tutorial.html
3.	https://www.google.co.in/books/edition/Learn_AngularJS_in_24_Hours/deX8DwAAQBAJ?hl=en&gbpv=1&dq=Angular+JS&printsec=frontcover

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5E002D	PEC-II (Computer Forensic)	3	0	0	3

Prerequisites for the course	
1.	Basic of Forensic.
2.	Fundamental of Computer Forensics.
3.	Understanding different Forensic Tools and Processing of Electronic Evidence.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/introduction-of-computer-forensics/
2.	https://mrcet.com/pdf/Lab%20Manuals/IT/R15A0533%20CF.pdf
3.	https://www.stannescet.ac.in/cms/staff/qbank/CSE/Notes/CS6004-CYBER%20FORENSICS-1800235714-CS6004%20UNIT%203.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will explain and properly document the process of digital forensics analysis.
2	CO2	Students will gain an understanding of the tradeoffs and differences between various forensic tools.
3	CO3	Students will be able to describe the representation and organization of data and metadata within modern computer systems.
4	CO4	Students will be able to create disk images, recover deleted files and extract hidden information.
5	CO5	Students will be introduced to the current research in computer forensics. This will encourage them to define research problems and develop effective solutions.

Syllabus:

Course Contents		Hours
Unit I	**Cyber Crime and computer crime:** Introduction to Digital Forensics, Definition and types of cybercrimes, electronic evidence and handling, electronic media, collection, searching and storage of	08

	electronic media, introduction to internet crimes, hacking and cracking, credit card and ATM frauds, web technology, cryptography, emerging digital crimes and modules.	
Unit II	**Cyber Laws:** Security Assurance, Security Laws, IPR , International Standards ,Security Audit , SSE-CMM/COBIT etc.	06
Unit III	**Computer Forensics:** Definition and Cardinal Rules, Data Acquisition and Authentication Process, Windows Systems-FAT12, FAT16, FAT32 and NTFS, UNIX file Systems, mac file systems, computer artifacts, Internet Artifacts, OS Artifacts and their forensic applications.	06
Unit IV	**Forensic Tools and Processing of Electronic Evidence:** Introduction to Forensic Tools, Usage of Slack space, tools for Disk Imaging, Data Recovery, Vulnerability Assessment Tools, Encase and FTK tools, Anti Forensics and probable counters, retrieving information.	08
Unit V	**Process of computer forensics and digital investigations** : Processing of digital evidence, digital images damaged SIM and data recovery, multimedia evidence, retrieving deleted data: desktops, laptops and mobiles, retrieving data from slack space, renamed file, ghosting, compressed files.	09

Text Books	
1.	The Third Edition of Internet Cryptography" by Richard E.Smith
2.	Computer Forensics by John R.Vacca

Reference Books	
1.	The Third Edition of Computer Forensics and Cyber Crime: An Introduction by Marjie T.Britz
2.	"Digital Forensics and Cyber Crime" by Joshua I James and Frank Breitinger
3.	"Practical Forensic Imaging" by Bruce Nikkel

Useful Links	
1.	https://www.google.co.in/books/edition/Computer_Forensics_I/8naLPQAACAAJ?hl=en
2.	https://www.google.co.in/books/edition/Digital_Forensics_and_Cyber_Crime/fiDXuEHFLhQC?hl=en&gbpv=1&dq=Computer+Forensic+notes&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS5M003**	**MDM-III (Prompt Engineering)**	**2**	**0**	**0**	**2**

Prerequisites for the course	
1.	Basic knowledge of Python programming.
2.	Introductory understanding of machine learning or AI concepts.

Prior Reading Material/useful links	
1	OpenAI Documentation
2	HuggingFace Transformers
3	Google AI Blog: Prompting Techniques

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will be able to Understand and apply basic and advanced prompt engineering techniques.
2	CO2	Students will be able to Integrate and utilize APIs to build AI applications.
3	CO3	Students will be able to Design and implement AI-powered chatbots and utilities.
4	CO4	Students will be able to Optimize generative AI applications for latency and cost.
5	CO5	Students will be able to Evaluate prompt performance and select appropriate fallback strategies.

Syllabus:

Course Contents		Hours
Unit I	**Foundations of Prompt Engineering:** Introduction to Prompt Engineering, Structure of a prompt, Instruction, Context, Input, Output, Zero-shot, One-shot, Few-shot prompting, Role prompting and system messages.	05
Unit II	**Advanced Prompt Engineering:** Chain-of-thought prompting, Temperature, top-p, and max tokens, Prompt tuning and embedding-based	05

	prompts, Persona creation through prompt control, Prompt design patterns (e.g., ReAct, Self-Ask).	
Unit III	**API Integration and Tools:** Open AI API: Authentication, completion, chat endpoints, Hugging Face Transformers API and hosted models, Lang Chain introduction: agents, chains, tools, Introduction to vector databases: FAISS, Chroma DB, Use of JSON output formatting and response parsing.	05
Unit IV	**Building AI Applications:** Designing Chat bots: State ful memory and context retention, Building Summarizers and Translators, Developing Personal AI Assistants, Integration with UI frameworks or apps (e.g., Streamlit, Flask), Mini project design and planning.	04
Unit V	**Evaluation & Optimization:** Prompt evaluation metrics, relevance, coherence, and accuracy, Latency measurement and performance profiling, Token usage and cost estimation, Fallback strategies and multi-model orchestration, Final project demo, feedback, and peer review.	05

	Text Books
1.	François Chollet, *Deep Learning with Python*, Manning Publications, 2nd Edition, 2021.
2.	Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana**,** *Practical Natural Language Processing*, O'Reilly Media, 1st Edition, 2020.
3.	Stuart Russell, Peter Norvig**,** *Artificial Intelligence: A Modern Approach*, Pearson, 4th Edition, 2020.
	Reference Books
1.	Denis Rothman**,** *Transformers for Natural Language Processing*, Packt Publishing, 2nd Edition, 2022.
2.	Tom Hope, Yehezkel S. Resheff, Itay Lieder**,** *Learning TensorFlow: A Guide to Building Deep Learning Systems*, O'Reilly Media, 1st Edition, 2017.
	Useful Link
1.	https://platform.openai.com/docs
2.	https://github.com/openai/openai-cookbook
3.	https://huggingface.co/docs
4.	https://docs.langchain.com

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5M004	MDM-IV (Ethics, Bias and Safety in LLM)	2	0	0	2

Prerequisites for the course	
1.	Basic understanding of Artificial Intelligence, Machine Learning concepts.
2.	Familiarity with Large Language Models (e.g., GPT, BERT).

Prior Reading Material/useful links	
1	OpenAI: https://openai.com/research
2	DeepMind Ethics & Society: https://deepmind.com/about/ethics-and-society
3	AI Now Institute Reports: https://ainowinstitute.org/reports.html
4	EU AI Act Overview: https://artificialintelligenceact.eu
5	Responsible AI at Microsoft: https://www.microsoft.com/en-us/ai/responsible-ai

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will be able to Analyze ethical challenges in LLM development and deployment.
2	CO2	Students will be able to Detect and mitigate bias in training datasets and model outputs.
3	CO3	Students will be able to Assess and improve the safety of LLM-based applications.
4	CO4	Students will be able to Understand legal and regulatory frameworks surrounding AI usage.
5	CO5	Students will be able to Evaluate and propose solutions to ethical controversies in LLMs using real-world case studies.

Syllabus:

Course Contents		Hours
Unit I	**Ethical Foundations in LLMs:** Introduction to AI ethics, Responsible AI principles, Ethics in NLP and LLMs: Accountability, transparency, and	05

	fairness, Ethical decision-making models, Introduction to fairness in machine learning.	
Unit II	**Privacy, Misinformation, and Hallucinations:**Data privacy concerns in LLM training and inference, Types and sources of misinformation and hallucinations, Mechanisms of hallucinations in generative models, Combatting fake news and deepfakes with AI.	05
Unit III	**Bias in Training Data and Outputs:**Sources of bias: gender, race, culture, Dataset auditing and de-biasing techniques, Evaluation metrics for bias and fairness, Societal impacts of biased outputs.	04
Unit IV	**Model Safety, Moderation, and Robustness:**Adversarial inputs and prompt injection, Jailbreaking LLMs and ethical concerns, Content filtering and moderation systems, Techniques for improving model safety and robustness.	05
Unit V	**Governance, Regulation, and Case Studies:**AI Governance: AI Act (EU), GDPR, OpenAI policy, and global perspectives, Responsible deployment strategies, Case Studies: Microsoft Tay, Meta Galactica, OpenAI ChatGPT, Analysis of mitigation strategies and lessons learned.	05

	Text Books
1.	Shannon Vallor, *"Technology and the Virtues: A Philosophical Guide to a Future Worth Wanting"*, Oxford University Press, 2016.
2.	Virginia Dignum, *"Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way"*, Springer, 2019.
	Reference Books
1.	Kate Crawford, *"Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence"*, Yale University Press, 2021.
2.	Sandra Wachter et al., *"Algorithmic Bias Detection and Mitigation: Best Practices and Policies"*, Oxford Internet Institute Whitepaper, 2020.
3.	Luciano Floridi, *"The Ethics of Artificial Intelligence"*, Oxford University Press, 2022.
	Useful Link
1.	https://arxiv.org – Search for keywords like "ethical LLM", "AI hallucinations", "LLM bias"
2.	https://huggingface.co/blog – Practical insights on AI deployment
3.	https://www.oecd.ai – OECD AI policy observatory
4.	https://futureoflife.org – AI safety and existential risk resources
5.	https://ai.facebook.com/blog – Meta AI's perspective on LLM ethics

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5O003	Open Elective-III (Advanced Business Application Programming)	2	0	0	2

Prerequisites for the course	
1.	Knowledge of Object-Oriented Programming.
2.	Knowledge of security protocols for user authentication.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/sap-advanced-business-application-programming-abap/
2.	https://www.techtarget.com/searchsap/definition/ABAP

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Apply advanced Object-Oriented Programming (OOP) principles for business applications.
2	CO2	Student will able to Build end-to-end web applications using modern back-end frameworks and front-end frameworks.
3	CO3	Student will able to Design and develop applications using monolithic, microservices, **and** event-driven architectures.
4	CO4	Student will able to Integrate security best practices, such as authentication, authorization, encryption, and prevent common vulnerabilities.
5	CO5	Student will able to Deploy and manage business applications in cloud environments.

Syllabus:

Course Contents		Hours
Unit I	**Advanced Programming Concepts and Object-Oriented Design**: Basic of Programming and Object-Oriented Design Focuses on advanced OOP principles, design patterns (Factory, Singleton, etc.), and data structures. Design scalable and reusable business logic, SOLID principles to improve software design.	06

Unit II	**Web Application Development and Frameworks:** backend development using frameworks like Spring Boot**,** ASP.NET Core**,** or Django**,** front-end development using React **or** Angular**.** full-stack applications by integrating front-end and back-end systems, managing state, and handling authentication.	05
Unit III	**Enterprise Application Architecture and Integration**: Explores enterprise architecture patterns, monolithic, microservices, and event-driven systems. external services, build scalable microservices, and work with distributed systems and APIs..	05
Unit IV	**Security and Compliance in Business Applications**: Web application security concepts, preventing SQL injection, XSS, and CSRF, role-based access control (RBAC), encryption, and comply with industry standards like GDPR and HIPAA.	05
Unit V	**Cloud Deployment, DevOps, and Business Application Maintenance**: Cloud computing concepts (AWS, Azure) and deployment strategies, Docker containerization and Kubernetes orchestration. continuous integration/deployment (CI/CD), monitoring, maintaining, and scaling applications in production environments..	05

Text Books	
1.	An Object-Oriented Approach to Programming Logic and Design, Joyce Farrell ·2012
2.	Java EE 7 Application Developer: Paper Code 1Z0-900, Manish Soni · 2024

Reference Books	
1.	British Qualifications - Page 642, 2003 · **Snippet view**
2.	Java by Dissection, Charlie McDowell, · **Ira Pohl2006**

Useful Links	
1.	https://cse.kiit.ac.in/wp-content/uploads/2024/01/Kareer-School.pdf
2.	https://www.cromacampus.com/blogs/sap-abap-course-syllabus-for-beginners/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5L004	Database Management Systems Lab	0	0	2	1

Prerequisites for the course	
1.	Understand Basics of Discrete Mathematics
2.	Fundamental of SQL.

Prior Reading Material/useful links	
1.	https://onlinecourses.nptel.ac.in/noc23_cs29/preview
2.	https://www.techtarget.com/searchdatamanagement/definition/database-management-system
3.	https://www.youtube.com/watch?v=c5HAwKX-suM

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to Apply the basic concepts of Database Systems and Applications.
2	CO2	Student will be able to Use the basics of SQL and construct queries using SQL in database creation and interaction.
3	CO3	Student will be able to Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4	CO4	Student will be able to analyse and Select storage and recovery techniques of database system.
5	CO5	Student will be able to Design the File Organization, Indexing and Hashing and exhibit the knowledge of transaction and concurrency control.

Course Contents List of Practical's		Hours
0	Introduction to Database Management Lab.	02
1	Creating tables, Renaming tables, Data constraints (Primary key, Foreign key, Not Null), Data insertion into a table.	02
2	Grouping data, aggregate functions, Oracle functions (mathematical, character functions).	02

3	Sub-queries, Set operations, Joins.	02
4	Creation of databases, writing SQL and PL/SQL queries to retrieve information from the databases.	02
5	Assignment on Triggers & Cursors.	02
6	Normal Forms: First, Second, Third and Boyce Codd Normal Forms.	02
7	Assignment in Design and Implementation of Database systems or packages for applications such as office automation, hotel management, hospital management.	02
8	Deployment of Forms, Reports Normalization, Query Processing Algorithms in the above application project.	02
9	Large objects – CLOB, NCLOB, BLOB and BFILE.	02
10	Distributed data base Management, creating web-page interfaces for database applications using servlet.	02

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Henry Korth, Abraham Silberschatz& S. Sudarshan, *Database System Concepts*, McGraw-Hill Publication, 6th Edition, 2011.
2.	Bipin Desai, An Introduction to Database System, West Publishing Company, College & School Division, 1990.
3.	RaghuRamakrishnan, Johannes Gehrke, *Database Management Systems*, McGraw-Hill Publication, 3rd Edition, 2003.

Reference Books	
1.	Joel Murach, *Murach's Oracle SQL and PL/SQL for Developers*, Mike Murach& Associates, 2nd Edition, 2014.
2.	Wiederhold, *Database Design*, McGraw-Hill Publication, 2nd Edition, 1983.
3.	Navathe, Fundamentals of Database System, Addison-Wesley Publication, 6 th Edition, 2012.

Useful Links	
1.	https://www.youtube.com/watch?v=T7AxM7Vqvaw
2.	https://www.guru99.com/what-is-dbms.html
3.	https://nptel.ac.in/courses/106105175

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5L005	**Machine Learning Lab**	0	0	2	1

Prerequisites for the course	
1.	Basics Knowledge of Computer programming.
2.	Introduction to Computer programming Lab.

Prior Reading Material/useful links	
1.	https://www.javatpoint.com/machine-learning
2.	https://www.tutorialspoint.com/machine_learning/index.htm
3.	https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will be able to choose the concepts of computational intelligence in Machine Learning for Designing.
2	CO2	Students will be able to classify the data based on Bayes Theorem and other techniques.
3	CO3	Students will be able to apply ANN Techniques to address the real time problems.
4	CO4	Students will be able to build new software for Real time problems using Reinforcement Learning.
5	CO5	Students will be able to design different Board positions and different applications by using Analytical Learning.

List of Practical's

Course Contents		Hours
0	Introduction of Machine Learning Lab.	2
1	Write a python program to import and export data using Pandas library functions	2
2	Demonstrate various data pre-processing techniques for a given dataset	2
3	Implement Dimensionality reduction using Principle Component Analysis (PCA) method.	2

4	Write a Python program to demonstrate various Data Visualization Techniques.	2
5	Implement Simple and Multiple Linear Regression Models.	2
6	Develop Logistic Regression Model for a given dataset.	2
7	Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.	2
8	Implement Naïve Bayes Classification in Python	2
9	Build KNN Classification model for a given dataset.	2
10	Write a python program to implement K-Means clustering Algorithm.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Tom M. Mitchell, "Machine Learning", McGraw-Hill, 2010
2.	Bishop, Christopher. Neural Networks for Pattern Recognition. New York, NY: Oxford University Press, 1995.
3.	Introduction to Machine Learning, Fourth Edition By Ethem Alpaydin-2020-MIT Press

Reference Books	
1.	Ethem Alpaydin, (2004) "Introduction to Machine Learning (Adaptive Computation and Machine Learning)", The MIT Press.
2.	T. astie, R. Tibshirani, J. H. Friedman, "The Elements of Statistical Learning", Springer(2nd ed.), 2009

Useful Links	
1.	https://www.jnit.org/wp-content/uploads/2020/04/Machine-Learning-Lab-Manual.pdf
2.	https://deepakdvallur.weebly.com/machine-learning-laboratory.html
3.	https://mrcet.com/pdf/Lab%20Manuals/MACHINE%20LEARNING%20LAB.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS5P003**	**Field Project/ Mini Project-III**	**0**	**0**	**2**	**1**

Prerequisites for the course	
1.	Understand the creation of new technologies or innovations.
2	Learning business opportunities.

Prior Reading Material/useful links	
1.	https://www.youtube.com/watch?v=p8e5ZEpRmx0
2.	https://www.youtube.com/watch?v=if_z7pMA85g
3.	https://www.youtube.com/watch?v=V5yv5TNpiLE

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will able to creation of new technologies or innovations.
2	CO2	Students will able to address complex problems or challenges
3	CO3	Students will able to develop new computational models or simulations that help advance their research.
4	CO4	Students will able to create new products or services that can be commercialized.
5	CO5	Students will able to impact various aspects of society, from improving productivity and efficiency to enhancing user experience and addressing social challenges.

This is continuous work to the Mini Project phase I. Every student will have to submit a completed report (1 copy) * of the project work. Report preparation guidelines should be followed as per given format. The students will prepare a power point presentation of the work. Panel of examiners comprising of guide, internal examiner, senior faculty, external examiner, etc. will assess the performance of the students considering their quality of work.

Mini Project Phase I
1. Group Formation.
2. Domain Selection.
3. Guide Allotment
4. Title Finalization.

5. Introductory Seminar
6. Progress Seminar-1-Problem Identification-Literature Survey
7. Progress Seminar-2-Work Flow
8. Fulfil Objective –I of Project Work
9. Finalization of project in all respect.
10. Synopsis Submission *(For guide, Personal copy, Departmental library.)
11. In a presentation, the students should focus to clarify problem definition and analysis of the problem.

Text Books	
1.	Planning and Implementing Your Final Year Project — with Success! By Mikael Berndtsson, Jörgen Hansson, B. Olsson, Björn Lundell · 2013
2.	Computer Science Project Work Principles and Pragmatics By Sally Fincher, Marian Petre, Martyn Clark · 2001
3.	Thesis Projects, A Guide for Students in Computer Science and Information Systems By Mikael Berndtsson · 2008

Reference Books	
1.	SOFTWARE ENGINEERING PROJECT MANAGEMENT By Bill Brykczynski, Richard D. Stutz · 2006
2.	Software Engineering Body of Knowledge By IEEE Computer Society · 2014
3.	Quality Software Project Management By Robert T. Futrell · 2002

Useful Links	
1.	https://www.google.co.in/books/edition/Project_Summaries/0ggTG_ya8acC?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&pg=PA78&printsec=frontcover
2.	https://www.google.co.in/books/edition/Real_World_Software_Projects_for_Compute/X6seEAAAQBAJ?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&printsec=frontcover
3.	https://www.logicraysacademy.com/blog/final-year-projects-for-cse/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5I001	Internship-I	0	0	4	2

Prerequisites for the course	
1.	Basics of Internet Programming like coding, designing, Testing.
2.	Fundamental Knowledge of Computing and Programming.

Prior Reading Material/useful links	
1.	https://www.bharathuniv.ac.in/downloads/csc/BCS7L2Web%20Technology%20Lab.pdf
2.	https://gnindia.dronacharya.info/CSE/Downloads/Labmanuals/Web-Tech-Lab Manual.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will be able to understand the appropriate workplace behaviours in a professional setting.
2	CO2	Students will be able to demonstrate content knowledge appropriate to job assignment.
3	CO3	Students will be able to exhibit evidence of increased content knowledge gained through practical experience.
4	CO4	Students will be able to explain how the internship placement site fits into their broader career field.
5	CO5	Students will be able to evaluate the internship experience in terms of their personal, educational and career needs..

Course Details:

Provides the student with an opportunity to gain knowledge and skills from a planned work experience in the student's chosen career field. In addition to meeting Core Learning Outcomes, jointly developed Specific Learning Outcomes are selected and evaluated by the Faculty Internship Advisor, Worksite Supervisor, and the student. Internship placements are directly related to the student's program of study and provide learning experiences not available in the classroom setting. Internships provide entry-level, career-related experience, and workplace competencies that employer's value when hiring new employees. Internships may also be used as an opportunity to explore career fields. Students must meet with the Internship & Apprenticeship Coordinator prior to registering. The purpose of the Internship Program is to provide each student practical experience in a standard work environment. The Internship & Apprenticeship Coordinator and Faculty Internship Advisor will assist students in making the job a valuable and productive experience. Success in this job will help ensure development of skills necessary for a lasting and rewarding career in the future.

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5J003	JDCAP III- Placement Preparation Training	2	0	0	Audit

Prerequisites for the course	
1	Basic arithmetic skills including addition, subtraction, multiplication, and division.
2	Understanding of fundamental mathematical terms such as factors, multiples, and fractions.
3	Basic knowledge of English language (reading, writing, speaking).

Prior Reading Material/useful links	
1.	https://www.learntheta.com/aptitude-preparation
2.	https://pdf.exampundit.in/quantitative-aptitude
3.	https://www.placementpreparation.io/blog/best-websites-to-learn-quantitative-aptitude/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to analyse mathematical concepts related to ratios, proportions, partnerships, and age problems to solve both theoretical and practical scenarios.
2	CO2	Student will able to evaluate profit and loss concepts to solve problems involving cost price, selling price, marked price, and discounts effectively.
3	CO3	Student will able to perform efficiently in personal interviews by applying etiquette, self-presentation, and common interview strategies.
4	CO4	Student will able to express opinions clearly in a group setting and to demonstrate teamwork, listening, and conflict resolution skills.
5	CO5	Student will able to develop arguments with evidence and examples and structure essays logically with a clear introduction, body, and conclusion.

Syllabus:

Course Contents		Hours
Unit I	**Mixture and Allegation, Time and work, Speed time and distance:** Problems on mixtures, Problems on allegations with two and three compounds. Problems on unitary methods, Problems on alternate days and wages, Problems on chain-rule, Problems on pipes and cisterns. Problems on speed-distance-	6

	time, Problems on average and relative speed, Problems on trains, Problems on boats and streams.	
Unit II	**Probability, Permutation and Combination, Data Interpretation:** Problems on coins, dice, leap year and non-leap year, Problems on deck of cards, balls, Problems on linear arrangement, Problems when repetitions are allowed, Problems on selections. DI on Tables, Column Graphs, Bar Graphs, Line Charts, Pie Chart, Cascade.	6
Unit III	**Part – A Resume & Cover Letter** 1. Basics of Resume Writing – Types, Structure, and ATS-friendly formatting. 2. Resume customization for different industries; Action verbs and quantifying achievements. 3. Cover letter essentials – Personalization, showcasing value, and avoiding clichés. **Part – B Verbal Ability** 1. Vocabulary enhancement – Synonyms, antonyms, idioms, collocations. 2. Sentence correction, para-jumbles, reading comprehension strategies.	6
Unit IV	**Part –A Essay Writing** 1: Types of essays – Analytical, persuasive, narrative. 2: Cohesion & coherence, avoiding plagiarism, writing with clarity. **Part- B Group Discussion** 1: Types of GDs – Topic-based, case study, abstract. 2: Strategies for entry, participation, and summarizing. **Part – C Personal Interview** 1: Common HR questions and STAR method for answering. 2: Handling stress interviews & behavioral questions. 3: Grooming, body language, and post-interview etiquette.	6

Text Books	
1.	Soft skills by Dr. K. Alex.
2.	Business communication by Meenakshi Raman and Sangeeta Sharma.
3.	Quantitative Aptitude and Reasoning by R. V. Praveen.

Reference Books	
1.	Analytical and Logical reasoning By Sijwali B S.
2.	Quantitative aptitude for Competitive examination By R S Agarwal.
3.	How to Prepare for Quantitative Aptitude By Arun Sharma.
4.	The 7 habits of highly effective people by Stephen R Covey.

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Useful Links	
1.	https://prepinsta.com /
2.	https://www.upgrade.com/
3.	https://www.javatpoint.com/

Sr. No.	Course Code	Course Name	Subject In charge Name
1	CS5T001	Database Management Systems	Prof. Mittal Patne
2	CS5T002	Machine learning	Prof. Aachal Wani
3	CS5T003	Internet of Things	Prof. Siddharth Ghosh
4	CS5E001A	PEC-I (Blockchain Technology)	Prof. Khushboo Bisen
5	CS5E001B	PEC-I (Cluster Computing)	Prof. Priyanka Patil
6	CS5E001C	PEC-I (Interactive Graphics)	Dr. Supriya Sawwashere
7	CS5E002A	PEC-II (Big Data Analytic Tools & Techniques)	Prof. Rasika Mohitkar
8	CS5E002C	PEC-II (React JS)	Mr. Sandip Zade
9	CS5E002D	PEC-II (Computer Forensic)	Prof. Yogita Maske
10	CS5M003	MDM-III (Prompt Engineering)	Prof. Karishma Bobde
11	CS5M004	MDM-III (Ethics, Bias and Safety in LLM)	Prof. Pratik Kambale
12	CS5O003	Open Elective –III (Advanced Business Application Programming)	Prof. Maresh Shingade
13	CS5L004	Database Management Systems Lab	Ms. Jagruti Thakre
14	CS5L005	Machine learning Lab	Mr. Om Butle
15	CS5P003	Field Project/ Mini Project-III	Prof. Anuja Ghasad
16	CS5I001	Internship-I	Mr. Sandip Zade
17	CS5T004	JDCAP-III: Placement Preparation Training	-

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